

Equipment Inspection Report

P-101 Crude Charge Pump — Report INS-2026-0412 — Inspection date 28 March 2026

SYNTHETIC DOCUMENT - created for a Smart India Hackathon prototype demonstration. Not real plant data. No sponsor information is used.

1. Inspection Details

Field	Detail
Equipment	P-101, crude charge pump
Work order	WO-26041
Reason	Mechanical seal leakage at gland
Inspector	K. Nair, Reliability Engineer
Downtime	17 hours

2. Findings

- The cartridge mechanical seal was found leaking at the gland face. Seal faces showed circumferential scoring consistent with dry running.
- The seal flush line strainer was found approximately 60 percent blocked with scale and particulate. Differential pressure across the filter was not recorded prior to shutdown.
- Measured flush differential at the point of isolation was 1.4 bar, against a required minimum of 3 bar above suction pressure.
- Bearing condition was satisfactory. Vibration readings taken before shutdown were within limits.
- Coupling alignment was checked and found within tolerance.

3. Assessment

The failure is attributed to inadequate seal flush caused by strainer fouling, rather than to the seal itself. This is the fourth leakage failure on this equipment, and the interval between failures has reduced from 187 days to 94 days. The pattern is consistent with a progressive restriction in the flush circuit.

4. Recommendations

1. Replace the mechanical seal with a like-for-like cartridge unit, part SEAL-682-C2.
2. Reduce the flush filter element change interval from 6 months to 3 months.
3. Record flush differential pressure on each shift, per HAZOP action A-04-02.
4. Review the upstream source of particulate in the flush liquid.

Approval for the above work is requested from the maintenance manager.